Ignacio D. Lopez-Miguel, M.Sc.

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Employment History

2020 – present **Formal Verification Engineer,** CERN *Geneva, Switzerland*
- Development and usage of the tool PLCverif to automatically formally verify PLC code.
- Eclipse ecosystem with Java and Xtext.
- Research work to optimize model checking algorithms for programs containing complex data types.

2017 – 2019 **Model Validation Specialist,** Deutsche Bank *Frankfurt am Main, Germany*
- Validation of Machine Learning models to estimate credit score performing different analyses, such as assumptions validation, sensitivity, robustness, and back-testing.
- Validation of Natural Language Processing model for credit score of enterprises.
- Development of Machine Learning challenger models.
- Participation in Data Science competitions.

2015 – 2017 **Quantitative Consultant,** Management Solutions *Madrid, Spain, and London, UK*
- Design and development of tools in VBA to automate processes.
- Analysis of Hadoop-based ecosystem to integrate it in a data lake.

2015 – 2015 **Research Engineer,** Ifak e.V *Magdeburg, Germany*
- Experimental research in order to characterize fluids via acoustic waves.

2014 – 2015 **LEGO Robotics Teacher,** San Jose School *Valladolid, Spain*
- Teaching students aged from 10 to 14 how to program and build LEGO robots.

Education

2019 – 2021 **M.Sc. Artificial Intelligence Research, Menendez Pelayo Univ. (UIMP).** GPA: *9.5/10.*
Thesis title: *Simplification of Numeric Variables for PLC Model Checking.*

2016 – present **B.Sc. Mathematics, National University of Distance Education (UNED).** GPA: *8.7/10.*

2015 – 2017 **M.Sc. Business Consulting, ICADE Business School.** GPA: *8.4/10.*
Thesis title: *Machine Learning Applied to Credit Scoring.*

2011 – 2015 **B.Sc. Ind. Electronics and Automation Eng., Valladolid University.** GPA: *8.7/10.*
Thesis title: *Guided acoustic wave's phase and group velocity at low frequencies in fluid filled elastic pipes.*

Research Publications

1. **Lopez-Miguel, Ignacio D.**, Fernandez Adiego, B., Tournier, J.-C., Rodriguez-Aguilar, J. A., & Blanco Viñuela, E. (2021). Simplification of Numeric Variables for PLC Model Checking. In *19th ACM-IEEE International Conference on Formal Methods and Models for System Design (MEMOCODE'21).*
2. Fernandez Adiego, B., **Lopez-Miguel, Ignacio D.**, & Tournier, J.-C. (2021). Applying model checking to highly-configurable safety critical software: the SPS-PPS PLC program. In *ICALEPCS'21.*
3. **Lopez-Miguel, Ignacio D.**, Tournier, J.-C., & Fernandez Adiego, B. (2021). PLCverif: status of a formal verification tool for Programmable Logic Controller. In *ICALEPCS'21.*
4. **Lopez-Miguel, Ignacio D.** (2021). Survey on Big Data Preprocessing Techniques. In *XOVETIC'21.*

Skills

Languages — Spanish (mother tongue), English (fluent), German (intermediate), French (basic).

Coding — Python, Java, C, R, SQL, VBA, LaTeX, NuSMV

References

Juan A. Rodriguez-Aguilar, Research Professor at Artificial Intelligence Research Institute (IIIA-CSIC)

Borja Fernandez Adiego, Automation Engineer at CERN

Jean-Charles Tournier, Software Engineer at CERN and Lecturer at EPFL